Online Appendix

Mapping O*NET Tasks to ISCO Occupations using Text Similarity

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Data and replication code: `https://doi.org/10.5281/zenodo.20359118`

A Parameter selection

Parameter values are selected through a multi-phase unsupervised search over blend weights that determine the share of each text source in the final query and target embeddings. The sweep is conducted on O*NET 29.2, which was the most recent release of the SOC 2018 taxonomy series at the time of parameter selection. Because the five blend weights control only how pre-computed component embeddings are combined—not any feature specific to a particular O*NET release—the identified values transfer unchanged to all 63 O*NET releases; this transferability is corroborated by the stable validation results across versions reported in Appendix C. The selection procedure is described here before the validation results (Appendices B–D), as the parameter selection is performed entirely on unsupervised criteria; validation results are examined afterwards as a consistency check only.

Search procedure

An adaptive iterative sweep is conducted over the five blend weights: w_{soc}, w_{dwa}, w_{isco}, $w_{\text{isco-task}}$, $w_{\text{occ}} \in [0, 1]$. All variants share pre-computed embeddings for the SOC occupation titles, ISCO-08 task descriptions, and ESCO occupation text and skill inventory; the variable cost is only the fast blend and FAISS retrieval step. One feasibility constraint is enforced: $w_{\text{soc}} + w_{\text{dwa}} \leq 0.90$, keeping at least 10% weight on the raw task text in the query vector.

Round 1 (initial grid). Five uniformly-spaced values are placed across each parameter's full range, yielding a $5^5 = 3{,}125$ candidate grid reduced to 1,250 feasible configurations after the query-weight constraint.

Rounds 2–6 (adaptive zoom). After each round all five parameters are re-gridded simultaneously using a trust-region strategy based on the current local step h and the current best value b for each parameter:

- *Local grid boundary, not global bound*: expand by one full step, targeting $[b - 2h,\ b - h,\ b,\ b + h,\ b + 2h]$ (clipped to $[0, 1]$).
- *True global bound*: one-sided refinement with a halved step $h' = \max(h/2,\ h_{\min})$, where the step-balance floor $h_{\min} = \max_p(h_p)/4$ prevents any one dimension from becoming much finer than the others.
- *Interior*: symmetric refinement with halved step h', placing five points at $[b - 2h',\ b - h',\ b,\ b + h',\ b + 2h']$.

The sweep will not converge if any parameter is at a grid boundary or if any parameter has not yet been refined below its initial step size, ensuring at least one interior-zoom pass for each dimension. If clipping at $[0, 1]$ reduces a parameter's grid to fewer than five distinct points, additional points are placed at successive multiples of the current step around the best value until five distinct points lie within bounds. All five parameters are re-gridded simultaneously after each round; no coordinate-wise or sequential updating is used. The sweep is fully deterministic: component embeddings are cached on disk and reloaded identically across runs, and the feasible configuration set is enumerated in a fixed iteration order. When multiple configurations share the maximum sweep score within a round, the configuration encountered first in that order is used as the reference for the next zoom; in the six rounds run here, ties of this kind did not arise. The sweep terminates when the improvement in the best sweep score of the current round over the global best from all preceding rounds falls below 0.0005, *and* every parameter's current best value is strictly interior to its grid, *and* every parameter's current step is strictly below its Round 1 value of 0.25. The threshold 0.0005 corresponds approximately to a 0.15 percentage-point change in coverage, which is below the noise floor of a single ISCO coverage step.

The six rounds yield 16,000 candidate specifications in total. The per-round configuration counts decrease as the grid concentrates around the optimum: 1,250 (Round 1), 3,000 (Round 2), 2,375 (Round 3), 3,125 (Round 4), 3,125 (Round 5), and 3,125 (Round 6).

Sweep score

Each candidate is scored by

$$\text{sweep\_score} = \frac{3 \cdot \text{cov} + 2 \cdot \text{sim} - 2 \cdot \text{overload} - 2 \cdot \text{gini}}{9},$$

where cov is the ISCO-08 coverage share (share of 4-digit ISCO-08 unit groups receiving at least one task assignment), sim is the mean cosine similarity of retained task–ISCO links, overload is the share of tasks assigned to overloaded ISCO groups, and gini is the Gini coefficient of the task-count distribution across ISCO groups (Table A1). The formula is applied identically in every round without re-normalisation, so scores are directly comparable across rounds.

An ISCO-08 unit group is classified as overloaded if its task count exceeds $T = \max(200, Q_{0.95})$, where $Q_{0.95}$ is the 95th percentile of the per-group task-count distribution for the configuration being evaluated. The absolute floor of 200 ensures the threshold is not trivially tight in configurations with high concentration. The overload penalty term in the sweep score penalises parameter configurations that produce excessive task concentration, steering selection away from weight combinations in which broad ISCO groups absorb semantically unrelated tasks. Among the 16,000 candidate configurations, the overload share ranges from 1.2% to 31.0% (std 2.9 pp), confirming that this criterion actively discriminates between parameter configurations.

Table A1: Sweep score components.

Metric	Weight	Direction
ISCO-08 coverage share	3	maximize
Mean cosine similarity retained	2	maximize
Share of tasks in overloaded ISCO group	2	minimize
Gini coefficient of task count per ISCO group	2	minimize

The 3-2-2-2 weight scheme reflects a priority ordering fixed *before* running the sweep and was not itself optimised. Coverage receives the highest weight because any uncovered ISCO-08 unit group represents a hard gap in the crosswalk: without at least one task assignment, no task-based exposure estimate can be produced for that occupation group. The specific value of 3 is the smallest integer strictly greater than the equal weight of 2 assigned to each other criterion; any value in $(2, \infty)$ achieves the same strict priority ordering, and 3 is the natural minimal integer choice. Mean cosine similarity receives weight 2 as a direct quality signal for retained links. Overload and Gini each receive penalty weight 2 as complementary concentration guards. The overload term detects configurations where any single group becomes a dominant absorber (task counts above the data-adaptive threshold T); the Gini term penalises uneven distribution across all groups. Together they steer the sweep away from configurations with pathological concentration, which arise at extreme parameter values and are characterised by lower mean similarity and up to 9–13 overloaded groups with task counts exceeding 700. The denominator 9 equals the sum of absolute weights $(3 + 2 + 2 + 2)$; the score therefore ranges from $-4/9 \approx -0.44$ (no coverage, no similarity, full overload, maximal Gini) to $5/9 \approx 0.56$ (full coverage, maximum similarity, zero overload, zero Gini). All findings about optimal weight values in this section are conditional on this objective function. Alternative weighting schemes were not tested; placing greater emphasis on similarity, for instance, would shift the identified optimum toward configurations with fewer but higher-quality links.

Selected configuration

The configuration with the highest sweep score across all 16,000 candidates is selected as the production specification: $w_{\text{soc}} = 0.675$, $w_{\text{dwa}} = 0.030$, $w_{\text{isco}} = 0.80$, $w_{\text{isco-task}} = 0.60$, $w_{\text{occ}} = 0.60$ (selection score 0.832; sweep metrics: coverage 100%, mean similarity 0.718, Gini 0.446, overloaded share 2.9%; see Table A2). The low overloaded share reflects structural SOC–ISCO-08 taxonomy asymmetry (see Appendix E) rather than a quality problem requiring pipeline correction: the two groups exceeding the concentration threshold (ISCO 2310, ISCO 1345) each draw over 75% of their tasks from a single SOC major group, confirming a structural rather than catch-all origin. Table A2 shows the top configurations by sweep score. The top candidates all achieve ISCO-08 coverage of 99.8–100% and span a narrow range of sweep scores, confirming that the crosswalk is insensitive to small perturbations in these weights under the sweep-score objective. This robustness is with respect to the unsupervised objective function and does not directly imply robustness of external validation performance.

Table A2: Top configurations by sweep score (selection score is a normalised composite used only for ranking within the sweep; raw sweep scores are comparable across rows).

Rank	Changed parameter	Value	Selection score	S3 coverage	S3 mean similarity	S3 Gini
1	baseline	baseline	0.832	1.000	0.718	0.446
2	multiple	w_dwa=0.0; w_isco_task=0.875	0.832	1.000	0.700	0.434
3	w_isco	0.4375	0.831	1.000	0.718	0.448
4	multiple	w_dwa=0.1875; w_isco=0.4688; w_isco_task=0.7812	0.831	1.000	0.725	0.452
5	multiple	w_dwa=0.0; w_isco_task=1.0	0.830	1.000	0.697	0.430
6	multiple	w_dwa=0.1875; w_isco=0.4062; w_isco_task=0.7812	0.829	1.000	0.726	0.453
7	multiple	w_soc_title=0.125; w_isco_task=1.0	0.828	1.000	0.695	0.436
8	multiple	w_occ=0.0; w_isco_task=0.75	0.828	1.000	0.717	0.444
9	multiple	w_dwa=0.0; w_occ=0.0; w_isco_task=0.875	0.827	1.000	0.698	0.431
10	multiple	w_occ=0.0; w_isco_task=0.875	0.827	1.000	0.716	0.441
11	w_isco_task	0.875	0.827	1.000	0.718	0.443
12	w_isco_task	0.875	0.827	1.000	0.718	0.443

The chosen configuration can be read directly from its weights. On the *query side*, the blend is constructed in two sequential steps. First, the raw task embedding and the DWA label embedding are combined: $\text{core} = (1 - w_{\text{dwa}}) \cdot \text{task} + w_{\text{dwa}} \cdot \text{DWA}$. Second, the SOC title embedding is blended in: $\text{query} = (1 - w_{\text{soc}}) \cdot \text{core} + w_{\text{soc}} \cdot \text{title}$. In the implementation, component vectors are L2-normalised before blending and the query vector is L2-normalised after each blend, so the weights should be interpreted as nominal blend weights rather than exact post-normalisation vector shares. Because the dials are applied sequentially, the effective contributions of each source are: SOC title 67.5% ($= w_{\text{soc}}$), raw task text 31.5% ($= (1 - w_{\text{soc}})(1 - w_{\text{dwa}})$), and DWA labels 1.0% ($= (1 - w_{\text{soc}}) \cdot w_{\text{dwa}}$). The SOC title provides the largest occupational context signal, with the raw task text contributing task-specific content, and DWA labels supplying a small complementary functional signal.

On the *target side*, $w_{\text{isco}} = 0.80$ allocates 80% of the target representation to ISCO-08 text and 20% to the ESCO component. Within the ISCO component, $w_{\text{isco-task}} = 0.60$ gives 60% weight to ISCO task-item descriptions versus 40% to the generic group description. Within the ESCO component, $w_{\text{occ}} = 0.60$ allocates 60% to occupation descriptions and 40% to the ESCO skill inventory. In nominal target-side weights, ISCO task items are the single largest contributor to the target embedding (48%), followed by ISCO information text (32%), ESCO occupation text (12%), and ESCO skill texts (8%). The target vector is L2-normalised before retrieval. Within the selected optimization framework, the official ISCO-08 task descriptions provide the dominant target-side anchor for matching O*NET tasks to ISCO-08 unit groups under the chosen objective.

Figures A1 and A2 show the sweep score surface for the target-side and query-side parameters respectively. Each cell reports the maximum sweep score across all configurations sharing that pair of parameter values (marginalising over the remaining parameters). The selected configuration is marked with a red star.

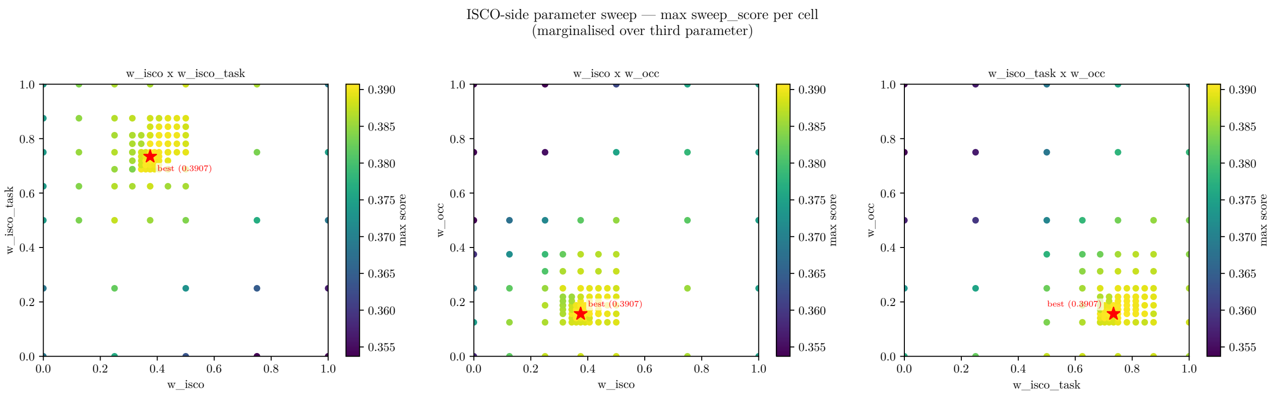

Figure A1: Target-side parameter sweep: maximum sweep score per cell for pairwise combinations of w_{isco}, $w_{\text{isco-task}}$, and w_{occ} (third parameter marginalised to its best-case value). Each cell therefore reports an upper bound on performance for that pair, not an average across all values of the remaining parameter. Red star marks the selected configuration.

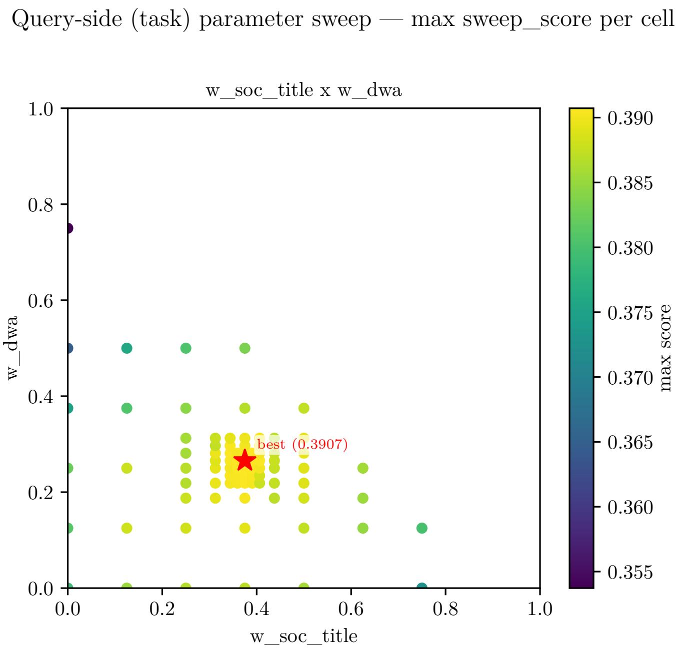

Figure A2: Query-side parameter sweep: maximum sweep score per cell for w_{soc} and w_{dwa} (target-side parameters marginalised to their best-case values; each cell reports an upper bound for that pair). Red star marks the selected configuration.

Findings

The selected DWA weight is near zero ($w_{\text{dwa}} = 0.030$, translating to $\approx 1.0\%$ of the query vector). The sweep score is identical whether $w_{\text{dwa}} = 0.030$ or $w_{\text{dwa}} = 0$ (both rank at 0.832), and the ablation study (Appendix G) confirms no measurable effect on validation agreement when DWA labels are removed entirely. Large DWA weights reduce performance, indicating that the DWA vocabulary adds little beyond the task statement and can dilute the stronger signals. The SOC title (67.5%) and raw task text (31.5%) dominate the query-side representation.

On the target side, ISCO-08 task descriptions dominate the optimal representation (effective weight 48% of the full target vector), with ISCO information text contributing 32% and ESCO occupation text and skill inventory accounting for the remaining 20%. The high weight on ISCO-08 text ($w_{\text{isco}} = 0.80$) reflects the discriminative power of official task descriptions, which directly describe the activities performed in each unit group. Combining ISCO-08 descriptions with ESCO occupation text improves retrieval over using either source alone.

Each task is assigned to a single ISCO-08 unit group. Allowing multiple assignments increases coverage but reduces interpretability and complicates aggregation of task-based measures. Configurations with high sweep scores cluster in a narrow region of the parameter space (see Table A2), indicating that the selected configuration is not a knife-edge optimum with respect to the unsupervised objective.

B Validation framework

Validation compares occupation-level assignments implied by the task-based mapping to institutional SOC–ISCO concordances constructed by statistical agencies and international organisations. Task-level assignments are first aggregated to the SOC occupation level. Each SOC occupation is assigned the ISCO-08 unit group receiving the largest number of task matches. Agreement is then evaluated relative to institutional SOC–ISCO mappings at both the four-digit unit-group level and the one-digit major-group level.

Multiple institutional concordances are used in order to avoid dependence on any single mapping source. Agreement is evaluated relative to both strict and lenient benchmarks, defined formally as follows. Let $\mathcal{C}(s)$ denote the set of institutional crosswalks available for SOC occupation s, and let $I_c(s)$ denote the set of ISCO-08 unit groups that crosswalk c assigns to occupation s. The *strict benchmark* accepts the pipeline's assignment $\hat{\iota}(s)$ only if it falls in every available crosswalk's set:

$$I^{\text{strict}}(s) = \bigcap_{c \in \mathcal{C}(s)} I_c(s).$$

The *lenient benchmark* accepts $\hat{\iota}(s)$ if it falls in at least one:

$$I^{\text{lenient}}(s) = \bigcup_{c \in \mathcal{C}(s)} I_c(s).$$

The four-digit in-set rate is the share of SOC occupations s for which $\hat{\iota}(s) \in I^{\bullet}(s)$. Results are similar across benchmark definitions, indicating that agreement rates are not driven by any single concordance.

Institutional crosswalks are used only for validation and do not influence construction of the task-level mapping.

C Full validation results

The pipeline was applied to all O*NET releases from v4.0 (2001) through v30.3 (2025), covering 63 releases in total across five SOC taxonomy generations (pre-2006, SOC 2006, SOC 2009, SOC 2010, SOC 2018). Table C1 reports chain crosswalk validation results for alternative benchmark definitions across three representative releases per major SOC taxonomy era: O*NET 25.1, 29.2, and 30.3 for the SOC 2018 generation (first, parameter-selection, and latest releases); and O*NET 15.1, 20.0, and 25.0 for the SOC 2010 generation (first, mid-era, and last releases). These two eras are shown because the institutional crosswalks available for validation differ between them: the ESCO–ONET-SOC chain crosswalks cover SOC 2018 (O*NET 25.1 onward), while the ESCO–O*NET and BLS–ISCO crosswalks cover SOC 2010. The main text reports results for O*NET 29.2 using the lenient benchmark, which balances measurement error across institutional concordances.

Table C1: Chain crosswalk validation: match rates across scenarios and representative O*NET releases.

Scenario	Cov. (%)	Exact (%)	Sub-major (%)	Major group (%)
*SOC 2018 — O*NET 25.1 (first SOC 2018 release)*				
A1 – ESCO-SOC18	99.9	81.5	89.7	94.0

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Table C1 – continued

Scenario	Cov. (%)	Exact (%)	Sub-major (%)	Major group (%)
A2 – SOC18-ESCO	98.7	73.7	84.1	88.9
A3 – Strict intersection	98.5	73.7	84.0	88.9
A4 – Lenient union	99.9	81.6	90.0	94.1
*SOC 2018 — O*NET 29.2 (release used for parameter selection)*				
A1 – ESCO-SOC18	99.9	81.5	89.7	94.1
A2 – SOC18-ESCO	98.7	73.8	84.3	89.0
A3 – Strict intersection	98.5	73.8	84.1	88.9
A4 – Lenient union	99.9	81.7	90.1	94.2
*SOC 2018 — O*NET 30.3 (latest release)*				
A1 – ESCO-SOC18	99.9	81.5	89.7	94.1
A2 – SOC18-ESCO	98.7	73.8	84.3	89.0
A3 – Strict intersection	98.5	73.8	84.1	88.9
A4 – Lenient union	99.9	81.7	90.1	94.2
*SOC 2010 — O*NET 15.1 (first SOC 2010 release)*				
B1 – ESCO-ONET-MHV[a]	79.1	63.1	76.1	83.1
B2 – SOC10-ISCO-BLS	99.9	57.9	70.5	77.7
B3 – Lenient union (MHV ∪ BLS)	100.0	63.4	76.7	82.7
B4 – Strict intersection (MHV ∩ BLS)	70.0	63.2	74.5	80.5
*SOC 2010 — O*NET 20.0 (mid-era)*				
B1 – ESCO-ONET-MHV[a]	79.3	62.8	76.3	83.2
B2 – SOC10-ISCO-BLS	99.9	57.6	70.4	77.7
B3 – Lenient union (MHV ∪ BLS)	100.0	63.1	76.7	82.8
B4 – Strict intersection (MHV ∩ BLS)	70.2	63.0	74.4	80.5
*SOC 2010 — O*NET 25.0 (last SOC 2010 release)*				
B1 – ESCO-ONET-MHV[a]	79.1	63.0	76.3	83.2
B2 – SOC10-ISCO-BLS	99.9	57.6	70.5	77.7
B3 – Lenient union (MHV ∪ BLS)	100.0	63.2	76.8	82.8
B4 – Strict intersection (MHV ∩ BLS)	70.0	63.2	74.6	80.6

[a]Less independent: derived via the same semantic similarity approach as the pipeline.

Confidence intervals (95% Wilson): For O*NET 29.2, scenario A4 (n = 18,755 tasks in crosswalk): exact [81.1%, 82.2%], sub-major [89.7%, 90.5%], major [93.9%, 94.5%]. For SOC 2010, scenario B3 ($n \approx$ 19,700): exact [±0.7 pp], major [±0.5 pp]. All intervals are within ±0.7 percentage points.

Table C2 shifts from task-level to occupation-level validation. Whereas Table C1 evaluates each task's ISCO assignment individually, Table C2 collapses the pipeline output to one ISCO per SOC occupation (the mode across tasks in that SOC) and then compares this implied SOC→ISCO mapping against institutional crosswalks. This complements Table C1 by checking whether the pipeline's aggregate occupational mapping is coherent with what expert-built crosswalks say, independently of the task-level scatter.

The table is restricted to the same six representative releases as Table C1. Two metrics are reported. *Pair precision* is the share of (SOC, ISCO) pairs implied by the pipeline that appear in the institutional crosswalk; *in-set rate* (4-digit, 2-digit, 1-digit) is the share of SOC occupations for which the pipeline's top-1 ISCO falls anywhere in the crosswalk's acceptable set at that level of the hierarchy.

Pair precision is structurally low: the pipeline assigns exactly one ISCO per SOC occupation, while institutional crosswalks typically list several acceptable ISCO codes per SOC. An occupation where the pipeline picks the correct ISCO from among five reference options contributes $1/5 = 0.20$ to pair precision but 1.00 to the in-set rate. In-set rate is therefore the appropriate primary accuracy measure; pair recall and F1 are included for completeness.

SOC 2010 versions (O*NET 15.1–25.0) are evaluated against the ESCO–O*NET and BLS–ISCO crosswalks; SOC 2018 versions (O*NET 25.1–30.3) are evaluated against the ESCO–ONET-SOC crosswalks. These are different instruments with different many-to-many cardinalities, so pair precision and in-set rates should not be compared directly across the two SOC eras.

Table C2: Occupation-level agreement with institutional crosswalks (same six releases as Table C1).

Dataset	Institutional crosswalk	Pair precision	Pair recall	Pair F1	In-set (4-digit)	In-set (2-digit)	In-set (1-digit)	Shared SOC n
O*NET 15.1	ESCO-ONET (MHV)	0.129	0.711	0.219	0.708	0.785	0.825	576
O*NET 15.1	BLS ISCO-SOC	0.147	0.713	0.243	0.650	0.721	0.773	774
O*NET 20.0	ESCO-ONET (MHV)	0.127	0.710	0.216	0.708	0.785	0.826	576
O*NET 20.0	BLS ISCO-SOC	0.145	0.715	0.241	0.652	0.722	0.775	774
O*NET 25.0	ESCO-ONET (MHV)	0.131	0.702	0.221	0.733	0.811	0.852	576
O*NET 25.0	BLS ISCO-SOC	0.150	0.713	0.249	0.678	0.756	0.804	774
O*NET 25.1	ESCO-ONET-SOC (SOC18)	0.340	0.468	0.394	0.868	0.910	0.944	797
O*NET 25.1	ONET-SOC-ESCO (SOC18)	0.246	0.615	0.351	0.811	0.861	0.897	784
O*NET 29.2	ESCO-ONET-SOC (SOC18)	0.343	0.464	0.394	0.868	0.911	0.944	797
O*NET 29.2	ONET-SOC-ESCO (SOC18)	0.248	0.608	0.352	0.812	0.862	0.898	784
O*NET 30.3	ESCO-ONET-SOC (SOC18)	0.343	0.464	0.394	0.867	0.910	0.942	797
O*NET 30.3	ONET-SOC-ESCO (SOC18)	0.247	0.608	0.352	0.811	0.861	0.897	784

Table C3 reports internal consistency between pairs of institutional crosswalks for the same SOC version. High within-version agreement between independent reference sources indicates that the institutional benchmarks themselves are coherent.

Table C3: Internal consistency between pairs of institutional crosswalks.

SOC version	Reference A	Reference B	Pair precision	Pair recall	Pair F1	In-set (4-digit)	In-set (2-digit)	In-set (1-digit)
SOC 2010	ESCO-ONET (MHV)	BLS ISCO-SOC	0.613	0.543	0.576	0.810	0.856	0.895
SOC 2018	ESCO-ONET-SOC (SOC18)	ONET-SOC-ESCO (SOC18)	0.544	0.988	0.701	0.598	0.690	0.738

Table C4 lists the top mismatches between the pipeline and institutional crosswalks for one representative release per SOC era: O*NET 25.0 against the BLS ISCO-SOC crosswalk, and O*NET 29.2 against the ESCO–ONET-SOC crosswalk. These examples should be read as suggestive rather than definitive: disagreement can originate from noise in the reference crosswalk rather than from the pipeline. A case in point is *Proofreaders and Copy Markers* (SOC 43-9081, O*NET 29.2): the pipeline assigns ISCO 4413 (*Coding, Proofreading and Related Clerks*), which directly matches the occupational tasks; the crosswalk maps this SOC to ISCO 2642 (*Journalists*), a pairing that is difficult to justify on task-content grounds. Similarly, *Survey Researchers* (SOC 19-3022) are assigned by the pipeline to ISCO 4227 (*Survey and Market Research Interviewers*), while the BLS crosswalk places them in ISCO 2120 (*Mathematicians, Actuaries and Statisticians*). In cases such as these the pipeline assignment is arguably more accurate than the reference, so the in-set rate provides a lower bound on true classification quality.

Table C4: Illustrative mismatches between the pipeline and institutional crosswalks (four examples per release; each row shows the top-1 ISCO from both available reference crosswalks).

SOC Occupation		Pipeline ISCO		Ref. 1 ISCO		Ref. 2 ISCO	
Code	Title	Code	Label	Code	Label	Code	Label
*O*NET 25.0 (SOC 2010): Ref. 1 = BLS ISCO-SOC; Ref. 2 = ESCO–O*NET*							
19-3022	Survey Researchers	4227	Survey and Market Research Interviewers	2120	Mathematicians, Actuaries and Statisticians	—	
11-3131	Training and Development Managers	2424	Training and Staff Development Professionals	1212	Human Resource Managers	2424	Training and Staff Development Professionals
41-3041	Travel Agents	4221	Travel Consultants and Clerks	3339	Business Services Agents Not Elsewhere Classified	4221	Travel Consultants and Clerks
13-1151	Training and Development Specialists	2424	Training and Staff Development Professionals	2356	Information Technology Trainers	2424	Training and Staff Development Professionals
*O*NET 29.2 (SOC 2018): Ref. 1 = ONET-SOC-ESCO; Ref. 2 = ESCO–ONET-SOC*							
19-3022	Survey Researchers	4227	Survey and Market Research Interviewers	2120	Mathematicians, Actuaries and Statisticians	2120	Mathematicians, Actuaries and Statisticians
11-3131	Training and Development Managers	2424	Training and Staff Development Professionals	1212	Human Resource Managers	2359	Teaching Professionals Not Elsewhere Classified
43-9081	Proofreaders and Copy Markers	4413	Coding, Proofreading and Related Clerks	2642	Journalists	1349	Professional Services Managers Not Elsewhere Classified
17-3031	Surveying and Mapping Technicians	2165	Cartographers and Surveyors	3112	Civil Engineering Technicians	2165	Cartographers and Surveyors

D Author annotation of O*NET 29.2 task assignments

To complement the automated chain-crosswalk validation, 108 O*NET 29.2 tasks were selected and annotated by the author. Tasks were drawn from 36 SOC occupations covering all nine ISCO major groups, with occupations chosen in order of combined US and Nordic employment within each group and tasks selected by O*NET importance score (top three per occupation). For each task the author assigned the most appropriate ISCO-08 four-digit unit group from a shortlist of plausible candidates, without reference to the pipeline's predictions.

The evaluation below serves as a post-hoc consistency check on the unsupervised config selection performed in Appendix A. The selected configuration ($w_{\text{soc}} = 0.675$, $w_{\text{dwa}} = 0.030$, $w_{\text{isco}} = 0.80$, $w_{\text{isco-task}} = 0.60$, $w_{\text{occ}} = 0.60$, max links= 1) achieves 54.6% exact four-digit agreement with the author annotations, rising to 71.3% at the two-digit sub-major level and 81.5% at the one-digit major-group level, at a mean cosine similarity of 0.699. Given the small annotation sample ($n = 108$), 95% Wilson confidence intervals are wide: exact [45.2%, 63.7%], sub-major [62.2%, 79.0%], major [73.1%, 87.7%]. These results should be interpreted as indicative rather than definitive. On the automated chain-crosswalk benchmark (lenient union A4), the selected configuration achieves 85.2% exact agreement on this subset (sub-major 97.2%, major-group 99.1%).

These results confirm that the semantic similarity approach produces assignments broadly consistent with expert judgement at the occupation-group level. The annotation was performed by the author (single annotator) and should be interpreted as an approximate consistency check rather than a definitive gold standard; the wide confidence intervals above reflect both the small sample size and the absence of inter-rater reliability evidence. The automated chain-crosswalk benchmark, which is based on institutional SOC–ISCO concordances, is the primary validation criterion.

E Coverage and example matches

The pipeline was applied to all 63 O*NET releases from v4.0 (2001) through v30.3 (2025). For releases prior to v20.1, O*NET distributed data in text format only; Excel format is available from v20.1 onward. Occupation titles are merged from `Occupation Data.txt` for all pre-v20.1 releases so that occupational context is preserved in the query embedding across the full version range. All parameters and threshold values are identical across versions; only the input data (task list and SOC taxonomy generation) differs.

For O*NET 29.2 (the release used for parameter selection), the mapping assigns tasks to 401 of the 436 ISCO-08 unit groups. Groups without task assignments typically lack a clear functional counterpart in U.S. occupational data; the most structurally notable case is ISCO 1113 (Traditional Chiefs and Heads of Village), which has no counterpart in U.S. occupational data and for which no official ISCO-08 task descriptions are available.

Coverage is broad across major occupational groups, including professional, technical, clerical, service, and manual occupations. Differences in task counts across ISCO-08 groups reflect differences in the level of detail in O*NET task descriptions and structural differences between U.S. and international occupational classifications.

Figure E1 shows how coverage and mean similarity evolve across the three pipeline stages across all O*NET releases. The retrieval stage (S1) achieves broad coverage at the cost of many candidates per task; subsequent filtering stages progressively tighten precision while maintaining near-complete coverage. Differences in task counts across ISCO-08 groups reflect structural differences between the SOC and ISCO-08 taxonomies rather than assignment errors; groups with the highest counts draw over 75% of their tasks from a single SOC major group (see Table E3).

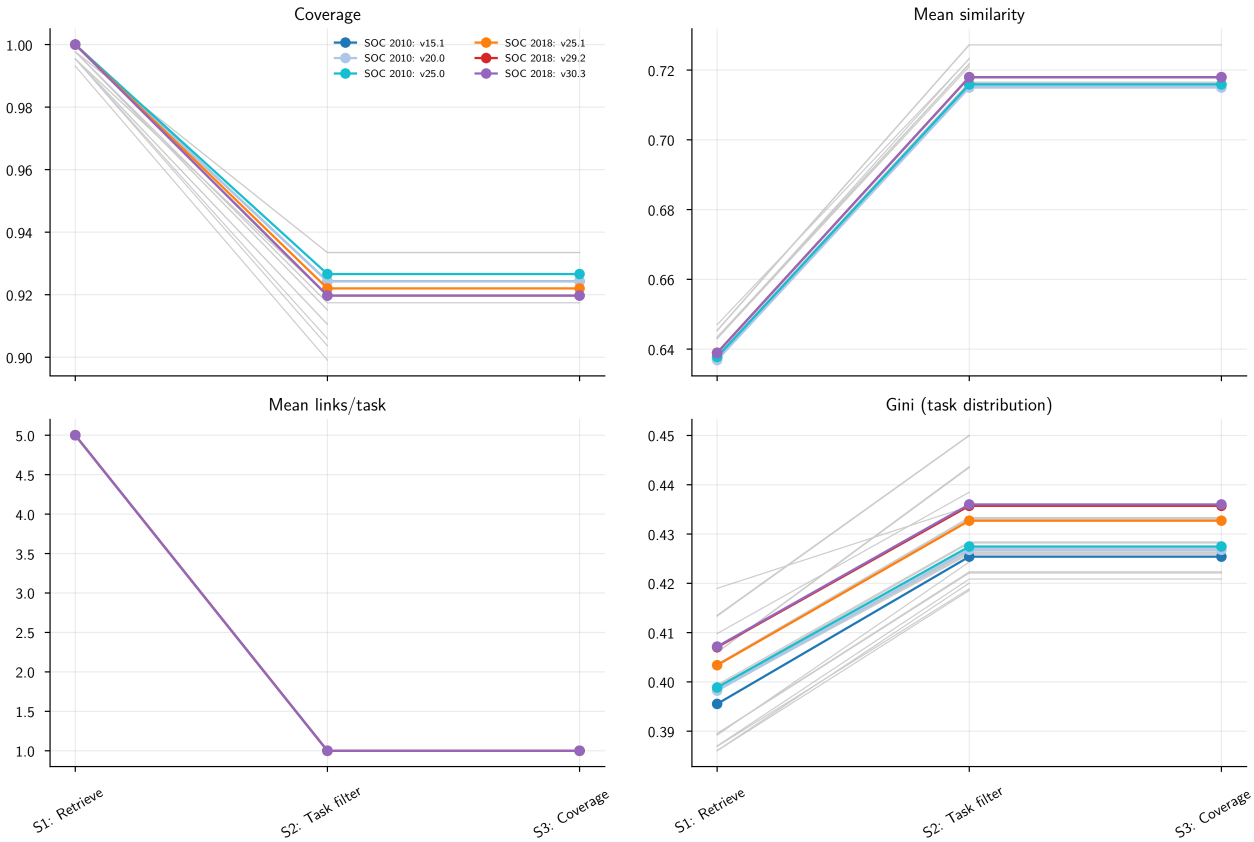

Figure E1: Pipeline metrics at each processing stage (S1–S3) across all O*NET releases (v4.0–v30.3).

Table E1 reports the corresponding numeric values for selected releases spanning the full version range.

Table E1: Pipeline metrics by stage and dataset.

Dataset	Stage	Cov.	Sim.	Links	Gini
O*NET 15.1	S1: Retrieve	1.000	0.638	5.000	0.396
O*NET 15.1	S2: Task filter	0.924	0.716	1.000	0.425
O*NET 15.1	S3: Coverage	0.924	0.716	1.000	0.425
O*NET 20.0	S1: Retrieve	1.000	0.637	5.000	0.398
O*NET 20.0	S2: Task filter	0.924	0.715	1.000	0.427
O*NET 20.0	S3: Coverage	0.924	0.715	1.000	0.427
O*NET 25.0	S1: Retrieve	1.000	0.638	5.000	0.399
O*NET 25.0	S2: Task filter	0.927	0.716	1.000	0.427
O*NET 25.0	S3: Coverage	0.927	0.716	1.000	0.427
O*NET 25.1	S1: Retrieve	1.000	0.639	5.000	0.403
O*NET 25.1	S2: Task filter	0.922	0.718	1.000	0.433
O*NET 25.1	S3: Coverage	0.922	0.718	1.000	0.433
O*NET 29.2	S1: Retrieve	1.000	0.639	5.000	0.407
O*NET 29.2	S2: Task filter	0.920	0.718	1.000	0.436
O*NET 29.2	S3: Coverage	0.920	0.718	1.000	0.436
O*NET 30.3	S1: Retrieve	1.000	0.639	5.000	0.407
O*NET 30.3	S2: Task filter	0.920	0.718	1.000	0.436
O*NET 30.3	S3: Coverage	0.920	0.718	1.000	0.436

Figure E2 compares final-stage (S3) summary metrics across all O*NET releases.

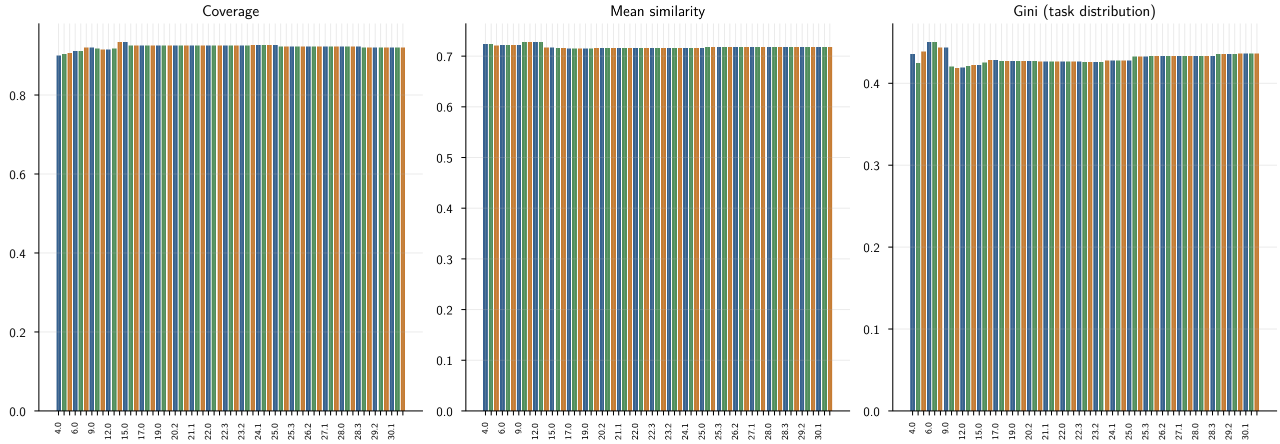

Figure E2: Final-stage (S3) metrics across all O*NET releases (v4.0–v30.3).

Table E2 summarises the S3 metrics numerically.

Table E2: Final-stage (S3) summary metrics across all O*NET releases (v4.0–v30.3).

Dataset	Cov.	Sim.	Links	Gini
15.100000	0.924	0.716	1.000	0.425
20.000000	0.924	0.715	1.000	0.427
25.000000	0.927	0.716	1.000	0.427
25.100000	0.922	0.718	1.000	0.433
29.200000	0.920	0.718	1.000	0.436
30.300000	0.920	0.718	1.000	0.436

Table E3 lists the ISCO-08 unit groups that attract the largest numbers of task assignments in the final output, shown for four representative releases. These groups are not catch-all categories; they are structurally dense nodes where the SOC taxonomy distinguishes many occupations that ISCO-08 unifies into a single unit group.

Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, for four representative releases.

Dataset	ISCO	Occupation label	Tasks
O*NET 20.0	1345	Education Managers	287
O*NET 20.0	2310	University and Higher Education	281
O*NET 20.0	7223	Metal Working Machine Tool Sette	223
O*NET 20.0	7233	Agricultural and Industrial Mach	200
O*NET 20.0	3131	Power Production Plant Operators	181
O*NET 20.0	3211	Medical Imaging and Therapeutic	180
O*NET 20.0	2351	Education Methods Specialists	178
O*NET 20.0	5329	Personal Care Workers in Health	171
O*NET 25.0	3211	Medical Imaging and Therapeutic	186
O*NET 25.0	5329	Personal Care Workers in Health	188
O*NET 25.0	2351	Education Methods Specialists	195
O*NET 25.0	2359	Teaching Professionals Not Elsew	186
O*NET 25.0	7233	Agricultural and Industrial Mach	207
O*NET 25.0	1345	Education Managers	271
O*NET 25.0	2310	University and Higher Education	289
O*NET 25.0	7223	Metal Working Machine Tool Sette	206
O*NET 29.2	3341	Office Supervisors	186
O*NET 29.2	2359	Teaching Professionals Not Elsew	188

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Table E3: ISCO-08 unit groups attracting the highest number of task assignments in the final output, for four representative releases.

Dataset	ISCO	Occupation label	Tasks
O*NET 29.2	7233	Agricultural and Industrial Mach	195
O*NET 29.2	7223	Metal Working Machine Tool Sette	191
O*NET 29.2	2212	Specialist Medical Practitioners	200
O*NET 29.2	1345	Education Managers	255
O*NET 29.2	2310	University and Higher Education	290
O*NET 29.2	2351	Education Methods Specialists	198
O*NET 30.3	2359	Teaching Professionals Not Elsew	188
O*NET 30.3	2310	University and Higher Education	290
O*NET 30.3	1345	Education Managers	257
O*NET 30.3	2212	Specialist Medical Practitioners	201
O*NET 30.3	2351	Education Methods Specialists	198
O*NET 30.3	7233	Agricultural and Industrial Mach	195
O*NET 30.3	7223	Metal Working Machine Tool Sette	188
O*NET 30.3	3341	Office Supervisors	186

Table E4 reports illustrative task assignments, one per ISCO major group, showing the full query-side inputs (task text, SOC occupation, DWA labels) alongside the pipeline assignment.

Table E4: Examples of task assignments (one per ISCO major group). O*NET task text, SOC occupation title, and the top two DWA activity labels constitute the query-side inputs; the ISCO-08 assignment is the pipeline output.

O*NET Task	SOC Occupation	DWA Labels (top 2)	ISCO-08 Assignment
Observe and monitor staff performance to ensure efficient operations and adherence to facility's policies and procedures	Lodging Managers	Evaluate employee performance.; Monitor activities of individuals to ensure safety or compliance with rules	Hotel Managers
Arrange public appearances, lectures, contests, or exhibits for clients to increase product or service awareness or to promote goodwill	Public Relations Specialists	Coordinate logistics for productions or events	Public Relations Professionals
Monitor animals recovering from surgery and notify veterinarians of any unusual changes or symptoms	Veterinary Assistants and Laboratory Animal Caretakers	Monitor patients to detect health problems	Veterinary Technicians and Assistants
Assemble and issue required documentation, such as tickets, travel insurance policies, or itineraries	Reservation and Transportation Ticket Agents and Travel Clerks	Compile data or documentation.; Make travel, accommodations, or entertainment arrangements for others	Travel Consultants and Clerks
Cut, trim and shape hair or hairpieces, based on customers' instructions, hair type, and facial features, using clippers, scissors, trimmers and razors	Hairdressers, Hairstylists, and Cosmetologists	Groom wigs or hairpieces.; Trim client hair	Hairdressers
Participate in the inspection, grading, sorting, storage, and post-harvest treatment of crops	Farmworkers and Laborers, Crop, Nursery, and Greenhouse	Sort forestry or agricultural materials	Field Crop and Vegetable Growers
Set up, program, operate, or tend computerized or manual woodworking machines, such as drill presses, lathes, shapers, routers, sanders, planers, or wood-nailing machines	Woodworking Machine Setters, Operators, and Tenders, Except Sawing	Operate woodworking equipment	Woodworking Machine Tool Setters and Operators
Start and control machines and equipment to wash, bleach, dye, or otherwise process and finish fabric, yarn, thread, or other textile goods	Textile Bleaching and Dyeing Machine Operators and Tenders	Operate garment treatment equipment	Bleaching, Dyeing and Fabric Cleaning Machine Operators
Clean and polish vehicle windows	Cleaners of Vehicles and Equipment	Clean vehicles or vehicle components	Vehicle Cleaners

F Embedding model robustness

The main specification uses `all-mpnet-base-v2` as the sentence encoder. To assess whether the results depend on this choice, we repeated the O*NET 29.2 mapping with two alternative encoders: `BAAI/bge-large-en-v1.5` (1024-dimensional) and `thenlper/gte-large` (1024-dimensional). All other parameters – including blend weights, similarity threshold, and margin – are identical to the production configuration.

Table F1 reports coverage, mean cosine similarity, and chain-crosswalk validation rates (scenario A4: lenient union of both SOC 2018 institutional crosswalks) for each encoder. Mean similarity values are not directly comparable across encoders because different models place embeddings at different scales within the unit sphere; the validation agreement rates are comparable.

Table F1: Embedding model comparison: O*NET 29.2, scenario A4 (SOC 2018 lenient union). Production weights applied unchanged to all three encoders.

Encoder		ISCO groups	Mean sim.	Cov. (%)	Exact (%)	Sub-major (%)
all-mpnet-base-v2	*(production)*	401	0.718	99.9	81.7	90.1
bge-large-en-v1.5		395	0.830	99.9	83.7	91.2
gte-large		390	0.936	99.9	84.7	91.8

Validation: O*NET 29.2, SOC 2018, scenario A4 (lenient union). All models use identical production weights. Mean similarity is no

The agreement rates are stable across encoders. Exact four-digit agreement (scenario A4) ranges from 81.7% (`all-mpnet-base-v2`) to 84.7% (`gte-large`), a difference of 3.0 percentage points. Major-group agreement ranges from 94.2% to 96.0% (1.8 pp). The two alternative encoders score marginally higher than the production model, reflecting their stronger performance on general semantic retrieval benchmarks; the production model was selected based on unsupervised sweep criteria rather than maximising validation agreement. ISCO-08 group coverage is consistent across encoders (390–401 of 436 groups). These results confirm that the core findings are not specific to the choice of `all-mpnet-base-v2`.

At the individual task level, however, approximately 25% of tasks receive a different four-digit ISCO-08 assignment depending on the encoder (MPNet vs. BGE: 73.4% exact agreement; MPNet vs. GTE: 75.0%; BGE vs. GTE: 87.6%). The two 1024-dimensional encoders are more similar to each other than either is to MPNet, reflecting differences in embedding geometry across model families. This task-level variation does not contradict the stable aggregate validation rates; it reflects genuine ambiguity in cases where multiple ISCO-08 unit groups are semantically plausible for a given task, and is consistent with the known difficulty of fine-grained cross-classification matching.

G Ablation study

To assess the contribution of each input component, we re-ran the pipeline on O*NET 29.2 with individual components removed, keeping all other parameters unchanged. Four variants are evaluated: removing the SOC occupation title from the query representation; removing the DWA activity labels; replacing the combined ISCO + ESCO target with pure ISCO-08 descriptions only; and using raw task text only (no occupational context at all).

Table G1 reports validation results for each variant against the lenient-union SOC 2018 benchmark (scenario A4).

Table G1: Ablation: validation rates when individual components are removed (O*NET 29.2, scenario A4, SOC 2018 lenient union).

Variant	Query side	Target side	ISCO groups	Mean sim.	Exact (%)	Sub-major (%)	Major (%)
Production (all components) *(production)*	SOC title + task + DWA	ISCO + ESCO	401	0.718	81.7	90.1	94.2
No SOC title	task + DWA only	ISCO + ESCO	426	0.608	43.6	64.2	78.4
No DWA labels	SOC title + task	ISCO + ESCO	401	0.717	81.7	90.1	94.1
No ESCO (ISCO-08 only)	SOC title + task + DWA	ISCO only	408	0.709	79.5	88.9	93.2
Task text only	task text only	ISCO + ESCO	426	0.603	43.5	64.0	78.2

Validation: O*NET 29.2, SOC 2018, scenario A4 (lenient union). All variants use identical threshold parameters ($\tau = 0.45$, $\delta = 0.03$).

Removing the SOC occupation title produces by far the largest drop: exact agreement falls from 81.7% to 43.6% (−38.1 pp), confirming that the SOC occupation title is the single most important signal for cross-taxonomy matching. Without it, the task text alone carries the full burden of disambiguation, and many semantically similar tasks spread across occupation groups receive the same ISCO-08 assignment. Removing the ESCO augmentation from the target side reduces exact agreement by 2.2 pp (79.5%), demonstrating that ESCO and ISCO-08 descriptions provide complementary information despite the modest weight difference. Removing DWA labels has no measurable effect (81.7%, identical to production), consistent with their small weight ($w_{\text{dwa}} = 0.030$). The task-text-only variant (43.5%) performs at the same level as removing the SOC title alone, confirming that the DWA labels contribute nothing once occupational context is absent.